

Lista de Exercício 7 - Coloração

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Exercício 01

The screenshot shows a C program in a code editor and its execution output in a console window. The program defines a 9x9 grid of integers and a function to display it with colored borders. The console output shows a 9x9 grid with a blue outer border, a red inner border, and a yellow center.

```
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>

#define dim 9

/*
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*/

int I[dim][dim] = {
    {1, 1, 1, 1, 1, 1, 1, 1, 1},
    {1, 0, 0, 0, 0, 0, 0, 0, 1},
    {1, 0, 4, 4, 4, 4, 4, 0, 1},
    {1, 0, 4, 15, 15, 15, 4, 0, 1},
    {1, 0, 4, 15, 14, 15, 4, 0, 1},
    {1, 0, 4, 15, 15, 15, 4, 0, 1},
    {1, 0, 4, 4, 4, 4, 4, 0, 1},
    {1, 0, 0, 0, 0, 0, 0, 0, 1},
    {1, 1, 1, 1, 1, 1, 1, 1, 1}
};

void exiba(int I[dim][dim]) {
    for(int i = -1; i < dim; i++) {
        _textcolor(8);
        for(int j = -1; j < dim; j++) {
            if( i < 0 && j < 0 ) {
                printf(" ");
            }
            else if( i < 0 ) {
                printf("%2d", j);
            }
            else if( j < 0 ) {
                printf("\n%2d", i);
            }
            else {
                _textcolor(I[i][j]);
                printf("%c", 219);
            }
        }
        _textcolor(8);
        printf("\n");
    }
}

int main(void) {
    exiba(I);
}
```

Console program output

```
0 1 2 3 4 5 6 7 8
0
1
2
3
4
5
6
7
8
Press any key to continue...
```

Exercício 02

```
Start Page x main.c x
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
#include "fila.h"
#define dim 9
#define cor(i,j) (i>=0 && i<dim && j>=0 && j<dim ? I[i][j] : -1)
#define par(i,j) ((i)*100+(j))
#define lin(p) ((p)/100)
#define col(p) ((p)%100)

/*
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*/

int I[dim][dim] = {
    {1, 1, 1, 1, 1, 1, 1, 1, 1},
    {1, 0, 0, 0, 0, 0, 0, 0, 1},
    {1, 0, 4, 4, 4, 4, 4, 0, 1},
    {1, 0, 4, 15, 15, 15, 4, 0, 1},
    {1, 0, 4, 14, 14, 14, 4, 0, 1},
    {1, 0, 4, 15, 15, 15, 4, 0, 1},
    {1, 0, 4, 4, 4, 4, 4, 0, 1},
    {1, 0, 0, 0, 0, 0, 0, 0, 1},
    {1, 1, 1, 1, 1, 1, 1, 1, 1}
};

void exiba(int I[dim][dim]) {
    system("cls");
    for(int i=-1; i<dim; i++) {
        _textcolor(8);
        for(int j=-1; j<dim; j++) {
            if(i<0 && j<0) printf(" ");
            else if(i<0) printf("%2d", j);
            else if(j<0) printf("\n%2d", i);
            else {
                _textcolor(I[i][j]);
                printf("%c%c", 219, 219);
            }
        }
        _textcolor(7);
    }
}

void colorir(int I[dim][dim], int i, int j, int n) {
    int a = I[i][j];
    if(a == n) return;
}
```

Console program output

```
0 1 2 3 4 5 6 7 8
0
1
2
3
4
5
6
7
8
Nova cor (0-15) ou -1 para sair? _
```

```
Start Page x main.c x
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
#include "fila.h"
#define dim 9
#define cor(i,j) (i>=0 && i<dim && j>=0 && j<dim ? I[i][j] : -1)
#define par(i,j) ((i)*100+(j))
#define lin(p) ((p)/100)
#define col(p) ((p)%100)

/*
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*/

int I[dim][dim] = {
    {1, 1, 1, 1, 1, 1, 1, 1, 1},
    {1, 0, 0, 0, 0, 0, 0, 0, 1},
    {1, 0, 4, 4, 4, 4, 4, 0, 1},
    {1, 0, 4, 15, 15, 15, 4, 0, 1},
    {1, 0, 4, 14, 14, 14, 4, 0, 1},
    {1, 0, 4, 15, 15, 15, 4, 0, 1},
    {1, 0, 4, 4, 4, 4, 4, 0, 1},
    {1, 0, 0, 0, 0, 0, 0, 0, 1},
    {1, 1, 1, 1, 1, 1, 1, 1, 1}
};

void exiba(int I[dim][dim]) {
    system("cls");
    for(int i=-1; i<dim; i++) {
        _textcolor(8);
        for(int j=-1; j<dim; j++) {
            if(i<0 && j<0) printf(" ");
            else if(i<0) printf("%2d", j);
            else if(j<0) printf("\n%2d", i);
            else {
                _textcolor(I[i][j]);
                printf("%c%c", 219, 219);
            }
        }
        _textcolor(7);
    }
}

void colorir(int I[dim][dim], int i, int j, int n) {
    int a = I[i][j];
    if(a == n) return;
}
```

Console program output

```
0 1 2 3 4 5 6 7 8
0
1
2
3
4
5
6
7
8
Nova cor (0-15) ou -1 para sair? _
```

Exercício 03

```
Start Page x main.c x
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
#include "fila.h"
#define dim 9
#define cor(i,j) (i>=0 && i<dim && j>=0 && j<dim ? I[i][j] : -1)
#define par(i,j) ((i)*100+(j))
#define lin(p) ((p)/100)
#define col(p) ((p)%100)

/*
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*/

int I[dim][dim];

void inicia(int I[dim][dim], char *s) {
    FILE *a = fopen(s, "r");
    if( !a ) {
        puts("arquivo imagem.txt nao encontrado!");
        printf("Certifique-se de que o arquivo esta na pasta do projeto.\n");
        abort();
    }
    for(int i=0; i<dim; i++) {
        for(int j=0; j<dim; j++) {
            if (fscanf(a, "%d", &I[i][j]) != 1) {
                I[i][j] = 0;
            }
        }
    }
    fclose(a);
}

void exiba(int I[dim][dim]) {
    system("cls");
    for(int i=-1; i<dim; i++) {
        _textcolor(8);
        for(int j=-1; j<dim; j++) {
            if( i<0 && j<0 ) printf(" ");
            else if( i<0 ) printf("%2d", j);
            else if( j<0 ) printf("\n%2d", i);
            else {
                _textcolor(I[i][j]);
                printf("%c%c", 219, 219);
            }
        }
    }
    _textcolor(7);
}

Console program output
0 1 2 3 4 5 6 7 8
0
1
2
3
4
5
6
7
8
Nova cor (0-15) ou -1 para sair? _
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```
Start Page x main.c x
#include <stdio.h>
#include <stdlib.h>
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#define dim 9
#define cor(i,j) (i>=0 && i<dim && j>=0 && j<dim ? I[i][j] : -1)
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/*
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int I[dim][dim];

void inicia(int I[dim][dim], char *s) {
    FILE *a = fopen(s, "r");
    if( !a ) {
        puts("arquivo imagem.txt nao encontrado!");
        printf("Certifique-se de que o arquivo esta na pasta do projeto.\n");
        abort();
    }
    for(int i=0; i<dim; i++) {
        for(int j=0; j<dim; j++) {
            if (fscanf(a, "%d", &I[i][j]) != 1) {
                I[i][j] = 0;
            }
        }
    }
    fclose(a);
}

void exiba(int I[dim][dim]) {
    system("cls");
    for(int i=-1; i<dim; i++) {
        _textcolor(8);
        for(int j=-1; j<dim; j++) {
            if( i<0 && j<0 ) printf(" ");
            else if( i<0 ) printf("%2d", j);
            else if( j<0 ) printf("\n%2d", i);
            else {
                _textcolor(I[i][j]);
                printf("%c%c", 219, 219);
            }
        }
    }
    _textcolor(7);
}

Console program output
0 1 2 3 4 5 6 7 8
0
1
2
3
4
5
6
7
8
Nova cor (0-15) ou -1 para sair? _
```

